लोक सेवा आयोग,
राजपत्राङ्कित तृतीय श्रेणी, नेपाल इन्जिनियरिङ्ग सेवा, सिभिल समूह, जेनरल उपसमूहको प्रतियोगितात्मक लिखित परीक्षा
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विषय: जेनरल सम्बन्धी |

Section-A (25 Marks)

1. **Draw a sketch of a suspension bridge and indicate its components. [10]**

Answer:

A suspension bridge primarily uses three types of cables to support and stabilize the bridge. Each cable has a specific function in transferring loads safely to the towers and anchorages.

The main components of suspension bridge are given below:

i. Main cable (Suspension Cable):

- ➢ It is the primary load-carrying cable of the bridge.
- ➢ It passes over the tops of the towers and is anchored firmly at both ends.
- ➢ Usually made of thousands of high-strength steel wires bundled together.
- ➢ Carries the load of the bridge deck and traffic through the suspenders.
- ➢ Transfers the load to the towers and anchor blocks.

Function:

- ➢ Supports the entire bridge deck.
- ➢ Transfers loads to towers and anchorages through tension.

ii. Hanger cables (Suspenders):

- ➢ These are **vertical steel cables or rods** connecting the bridge deck to the main cable.
- ➢ Placed at regular intervals along the span.

Function:

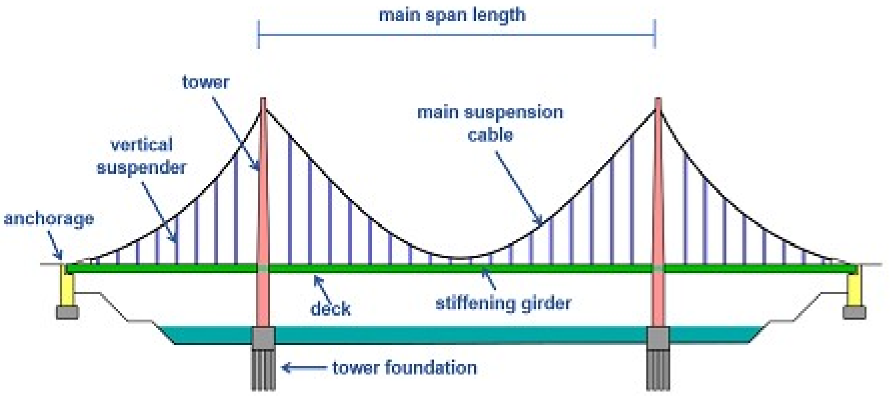
- ➢ Transfer the weight of the bridge deck and moving vehicles to the main cable.
- ➢ Keep the bridge deck suspended and properly aligned.

iii. Anchor cables:

- ➢ These are the portions of the main cable extending from the towers to the anchor blocks.
- ➢ They are firmly fixed into massive concrete anchorages.

Function:

- ➢ Balance the tension in the main cable.
- ➢ Transfer tensile forces safely to the ground through anchor blocks.
- ➢ Prevent the bridge from collapsing toward the center span.



In a suspension bridge, the load from vehicles and the bridge itself is transferred to the ground through a sequence of structural members.

Load Transfer Mechanism in flowchart diagram:

Traffic Load + Self-weight of Deck
↓
Bridge Deck
↓
Hanger Cables (Suspenders)
↓
Main Cables
↓
Towers (Pylons)
↓
Anchor Blocks
↓
Foundations
↓
Ground (Soil/Rock)

- **Bridge deck:**
  It carries traffic loads and transfers them to the hanger cables.
- **Hanger cables:**
  Hangar cables also known as suspenders carry tension.
- **Main cables:**
  Main cables also called suspension cables also carry tension.
- **Towers:**
  They carry compression.
- **Anchor blocks:**
  These blocks resist tension and prevent the main cables from pulling inward.
- **Foundations:**
  Foundation of the bridge finally distributes all forces safely to the supporting soil or rock.

2. **Calculate the stopping sight distance for the following 3 cases on a highway for a design speed of 80 kmph. Reaction time is 2 secs and friction coefficient are 0.4. [10]**
   a. When grade is 4% descending
   b. When grade is 3% ascending
   c. when road is flat

Answer:
Stopping Sight Distance (SSD) is calculated as:

$$SSD = 0.278\,V\,t + \frac{V^2}{254(f \pm G)}$$

$$Where:$$

$$V = Design \ speed = 80 \ km/h$$

$$t = Reaction \ time = 2 \ s$$

$$f = Coefficient \ of \ friction = 0.4$$

$$G = Gradient$$

$$+ \ \ for \ \ ascending \ \ grade$$

$$- \ \ for \ \ descending \ \ grade$$

Reaction time $= 0.278 \times 80 \times 2 = 44.48\, m$

a. When grade is 4% descending:

$$G = \frac{4}{100} = 0.04$$

Braking distance $= \dfrac{80^2}{254\left(0.4 - 0.04\right)} = \dfrac{6400}{254 \times 0.36} = \dfrac{6400}{91.44} = 69.99 \approx 70.0\,\mathrm{m}$

$$Therefore\,,$$

$$SSD = 44.48 + 69.99 = 114.47\,\mathrm{m} \approx 114.5\,\mathrm{m}$$

b. when grade is 3% ascending:

$$G = \frac{3}{100} = 0.03$$

$$Braking \ \ distance:$$

$$= \frac{80^2}{254\left(0.4 + 0.03\right)} = \frac{6400}{254 \times 0.43} = \frac{6400}{109.22} = 58.60\,\mathrm{m}$$

$$Therefore\,,$$

$$SSD = 44.48 + 58.60$$

$$= 103.08\,\mathrm{m} \approx 103.1\,\mathrm{m}$$

c. when road is flat:

Here, G=0

Braking distance $= \dfrac{80^2}{254 \times 0.4} = \dfrac{6400}{101.6} = 62.99 \approx 63.0\,\mathrm{m}$

$$Therefore\,,$$

$$SSD = 44.48 + 62.99$$

$$= 107.47\, m \approx 107.5\, m$$

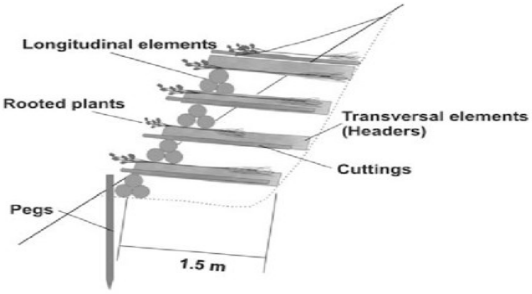
3. **Explain the use of bio engineering in Hill Roads and its advantages compared to other method. [10]**

Answer:

Stabilization of slope using vegetative systems with small civil engineering structures is known as bio-engineering. It combines biological measures (plants) with civil engineering works to improve slope stability in an environmentally friendly and cost-effective manner.

Example:

i. Toe wall with bamboo plantation:



ii. Jute net with grass seedling:

Principle of bio-engineering:

- The main concept of bio-engineering is the interaction between vegetative systems and small civil engineering structure.
- Plant roots bind soil particles and increase shear strength.
- Vegetation reduces the impact of rainfall and surface runoff.
- Plants absorb soil moisture, lowering pore water pressure.
- Natural cover protects slopes from erosion and weathering.

Use of bio-engineering in hill roads:

i. It stabilizes cut and fill slopes by reinforcing the soil with plant roots.
ii. It reduces soil erosion caused by rainfall and surface runoff.
iii. Bio-engineering also prevents shallow landslides and slope failures.
iv. It protects road embankments, drainage channels, and retaining structures.
v. It also improves slope drainage by increasing infiltration and reducing runoff velocity.
vi. Bio-engineering also restores vegetation cover and enhances the surrounding environment.

Application of Bio-engineering helps to stabilize slopes, control erosion, and reduce landslide risks. It is widely used on hilly roads in Nepal because it is economical, environment-friendly and sustainable.

The advantages of bio-engineering compared to other methods are listed below:

i. Bio-engineering requires low construction and maintenance cost in comparison to conventional methods.
ii. It is environment-friendly and aesthetically pleasing in comparison to other methods.
iii. Bio-engineering improves bio-diversity and landscape while other methods have little ecological benefits.
iv. This method is flexible and adapts to minor ground movement while other conventional methods are rigid and may fail due to settlement.
v. The method utilizes locally available materials and labor.
vi. Bio-engineering helps in environmental enhancement in comparison to other methods.

4. **Briefly explain the following: [4+3+3]**
a. Solid waste management
b. Source of water supply, their selection and management
c. Sources of pollution for air, water and land
Answer:

a. **Solid Waste management:**

- The systematic process of generation, segregation, collection, storage, transportation, processing, treatment, recycling, and final disposal of solid waste without directly or indirectly impacting on the environment, society and public health is known as Solid waste management.
- It can be classified into five main stages which is also referred as the functional elements of solid waste management. They are:
  i. Onsite handling, storage and processing
  ii. Collection
  iii. Transfer and transport
  iv. Resource recovery and processing
  v. Disposal

- Importance of solid waste management:
  i. It helps to make surrounding clean and prevent the spread of harmful diseases and bacteria.
  ii. It reduces the risk of communicable diseases.
  iii. It prevents the pollution of groundwater and surface water.
  iv. It prevents the pollution of land by efficient solid waste management.
  v. It utilizes the nitrogenous matter present in sewage as fertilizers.
  vi. It prevents the odor nuisance, sewage sickness etc.
- There are five principles of sloid waste management.
  i. Refuse
  ii. Reduce
  iii. Reuse
  iv. Recycle
  v. Recover

b. Sources of water supply, their selection and management.

- Any natural or artificial source from which water can be utilized for domestic, industrial, agricultural, recreation and other purposes is known as sources of water supply.
- Most of the water obtained through precipitation is retained in surface depressions, carried away as surface runoff in natural streams or rivers and percolates into the ground which joins groundwater.
- Hence, sources of water can be broadly classified into following types. They are:
  i. Surface water source:
    - Water available on the ground surface is known as surface source.
    - The quality and quantity of water available on the surface source depends on the rainfall patterns, climatic and geological factors.
    - Various forms of surface sources of water are:
      - ✓ Lakes and ponds
      - ✓ Stream or rivers
      - ✓ Storage or impounded reservoirs

  ii. Groundwater source:
    - Water that exists below the ground surface is termed as groundwater source.
    - The quality of groundwater is generally good due to natural infiltration as it infiltrates deep into the soil percolating into deeper strata.
    - Groundwater acquires its chemical characteristics from surface water that percolates into the ground.
    - Various forms of the ground sources are as follows:
      - ✓ Springs
      - ✓ Infiltration galleries
      - ✓ Infiltration wells
      - ✓ Wells and tube wells

Selection of water source:

The major factors considered while selecting the sources of water supply are as follows:

i. Quality of water:
   - The source should be able to supply a sufficient quantity of water to meet the various demands of the city during the entire design period.
   - Water availability in the source may vary with the season to season so the quantity of water available in dry season should be adequate to meet the water demand, i.e., safe demand.

ii. Quantity of water:
   - The source should have safe wholesome, free from pollution of any kind and other undesirable impurities.
   - The impurities present in the water should be as less as possible and these should be removed easily and cheaply.

iii. Location:
   - As far as possible source of water should be locating near to the community and it should be situated in an elevated area.
   - This will reduce the length of pipe and water from the source would flow by gravity and hence the cost of the project reduced.

Effective management of water sources ensures sustainable and safe water supply through:

- Watershed and catchment protection to reduce erosion and contamination.
- Regular monitoring of water quantity and quality.
- Pollution control by regulating waste disposal and industrial discharge.
- Groundwater recharge through rainwater harvesting and recharge structures.
- Maintenance of intake structures, wells, reservoirs, and pipelines.
- Community participation in operation, maintenance, and protection.

c. Sources of pollution for air, water and land

Pollution is the introduction of substances or energy into the natural environment in amounts or concentrations that can be harmful for humans, animals, and plants.

Sources of air pollution:

Air pollution is the presence of substances in the air that are harmful to humans, other living beings or the environment. Pollutants can be gases like carbon dioxide, methane etc. or small particles like dust and soot. Both outdoor and indoor air can be polluted. The sources of air pollution are:

- Vehicular emissions (CO, NOx, particulate matter)
- Industrial emissions (SO_2, NOx, smoke)
- Thermal power plants
- Forest fire
- Burning of fossil fuels (coal, diesel, petrol)
- Open burning of solid waste and agricultural residues
- Construction and mining activities
- Brick kilns and cement industries
- Household cooking using biomass fuels.

Sources of water pollution:

Water pollution also known as aquatic pollution is the contamination of water bodies which alters the physical, chemical and biological composition of water. The sources of water pollution are as follows:

- Sewage and wastewater
- Household detergents and chemicals
- Industrial effluents, Toxic chemicals and dyes
- Heavy metals (lead, mercury, cadmium)

- Fertilizers (nitrates and phosphates)
- Pesticides and herbicides
- Animal waste
- Oil spills and landfill leachate
- Mining activities
- Hospital and pharmaceutical waste
- Urban stormwater runoff

Sources of land pollution:
The unwanted addition or contamination of the land which alters the physical, chemical and biological composition of water is called land pollution. The sources of land pollution are:

- Open dumping of solid waste
- Improper landfill management
- Hazardous industrial waste
- Mining and quarrying waste
- Excessive use of fertilizers, pesticides and insecticides
- Biomedical waste
- E-waste (electronic waste)
- Construction and demolition debris
- Oil leakage from vehicles and machinery

5. **a. How the operation and maintenance of water supply and sewerage system can be improved? [5]**
b. Community-based water supply and sewerage systems are feasible and sustainable in Nepal. Justify this statement. [5]

Answer:
a. For the continuous, safe, efficient, and sustainable functioning of water supply and sewerage systems, proper operation and maintenance operations must be carried out. Proper O&M improves service quality, reduces system failures, and extends the life of infrastructure. The operation and maintenance of water supply and sewerage systems can be improved by:

i. Regular inspection and monitoring of pipelines, pumps, valves, reservoirs regularly.
ii. Repairing of leakage and damaged components of the system before major failure occurs.
iii. Reduce Non-Revenue water through metering and pressure management.
iv. Replacement of old and corroded pipelines and other appurtances.
v. Operation of treatment units according to design standards.
vi. Conduction of regular training to the operation personnel regarding operation, maintenance, safety of the water supply systems.
vii. Allocation of sufficient budget for routine maintenance and spare parts.
viii. Installation of flow meters, pressure gauges and SCADA systems for real-time monitoring.

b. In a community-based water supply and sewerage system local users actively participate in the planning, design, construction, operation, maintenance, and management of the system through user committees or Water Users and Sanitation Committees (WUSCs). Nepal has successfully implemented this approach, especially in rural areas due to easy availability of local raw materials required for construction activities and develop the sense of ownership ensuring long-term utilization of the infrastructure. Community-based water supply and sewerage systems are feasible and sustainable in Nepal which can be justified by following:

i. Strong community participation ensures contribution of local people leading to better protection of the system.
ii. Local management such as user's committee carries out quick repairs of leaks and other fault in the system.
iii. The revenue generated from the system can be utilized for operation, maintenance, repairs and future improvements.
iv. Local raw materials and appropriate technologies can be utilized at lower budget ensuring effectiveness of the project.
v. Encouragement for the participation of women, marginalized groups and local stakeholders in major decision-making.

vi. Promotion of transparency, accountability and equitable water distribution.
vii. Promotes efficient water use and sanitation practices.

Thousands of rural water supply schemes have been successfully managed by communities with technical support from the government and development partners. Community management has increased access to safe drinking water and sanitation in many rural municipalities.

6. **Write short noted on:**
a. Alternative Energy system in Nepal [5]
b. River diversion works [5]

Answer:

a. Alternate Energy system in Nepal:

It refers to the energy obtained from renewable and environmentally friendly sources. In Nepal, alternative energy plays a vital role in providing clean, reliable, and affordable energy, especially in rural and remote areas. As Nepal is a rich country in terms of natural resources, there is a humungous scope of alternate energy generation. There is an independent government institution called Alternative Energy Promotion Centre (AEPC) under Ministry of Energy, Water Resources and Irrigation. AEPC supports technologies such as solar, biogas, small hydro, improved cookstoves, and other decentralized renewable energy systems. Its goal is to improve energy access, especially in rural areas, while reducing environmental impact.

The major alternative energy systems in Nepal are listed below:

i. Hydropower:
- It is the electricity generated by utilizing the energy of flowing rivers.
- Globally, hydropower covers about 14.3% of total electricity generation.
- Nepal has more than 6000 big and small rivers which has net electricity generation capacity of 83000 MW making it a second richest country in water resources after Brazil.
- Due to steep topography that provides the required head for power generation and availability of fast flowing rivers in Nepal, hydropower is most suitable option of alternate energy system.

ii. Solar energy:
- It is the electricity generated by utilizing the sunlight.
- Powered by the photovoltaic effect (using silicon solar cells) or solar-thermal technology, it provides a renewable, clean, and increasingly cost-effective way to power homes, businesses, and entire utility grids.
- Solar energy covers about 10% of global electricity demand.
- It is ideal for remote mountain regions with high solar radiation where other alternate source of energy is difficult to supply or are expensive.

iii. Biogas:
- Biogas is a combustible gas produced by the breakdown of organic materials such as agricultural residues, animal manure, food waste, and sewage by microorganisms in an oxygen-free environment
- Biogas covers a very small fraction of total national energy consumption, accounting for less than 1%.
- It is very popular in rural households where agricultural wastages such as cow dungs can be utilized to produce cooking gas.
- Nepal has successfully generated and utilized locally produced green hydrogen to power hydrogen cookstoves of which the pilot facility was officially inaugurated in Municipality, Baglung, on 15th May 2026.

iv. Wind energy:
- Wind energy is a renewable power source that utilizes the kinetic energy of moving air to generate electricity.

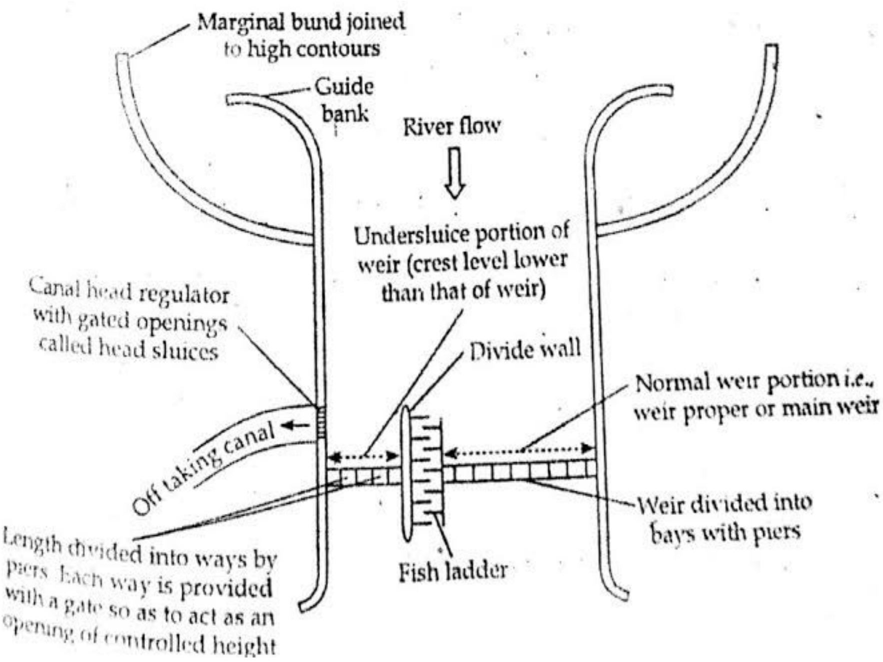
➢ Potential of wind energy exists in the district of Nepal with the presence of high velocity wind such as Mustang, Palpa, Dolpa, and other high-wind regions.
➢ It is currently limited in application.

Importance of alternate energy sources:

➢ Reduces dependence on imported fossil fuels.
➢ Provides electricity to remote and off-grid communities.
➢ Reduces greenhouse gas emissions and air pollution.

b**. River diversion works:**

The temporary or permanent hydraulic structures constructed to divert the flow of a river away from the construction site so that work on dams, bridges, hydropower projects, and other river structures can be carried out safely in dry conditions is known as river diversion works.



Types of river diversion works:

i. **Cofferdam:**
   ➢ It is a temporary watertight structure built around the construction site to keep the water out for the construction.
   ➢ Types of coffer dam:

      - Earth-fill cofferdam
      - Rock-fill cofferdam
      - Sheet pile cofferdam
      - Cellular cofferdam

ii. **Diversion tunnel:**
   ➢ It is a tunnel excavated through the river bank to carry river flow around the construction site,
   ➢ It is commonly constructed in hydropower and dam projects.

iii. **Diversion channel:**
   ➢ It is the artificial channel constructed to bypass the river flow through the work area.
   ➢ It is suitable where sufficient land is available.

iv. **Weir or barrage:**
   ➢ Weir is an impervious barrier constructed across a river to raise the upstream water level, facilitating the diversion of water into an intake.

- Barrage is similar to a weir but equipped with adjustable gates along its entire length. This provides better control over the pond level.

River diversion works are widely used in Nepal's hydropower sector and irrigation projects. Most of the large projects such as Upper Tamakoshi, Middle Bhotekoshi, Kali Gandaki A, and other storage or run-of-river schemes utilized cofferdams and diversion tunnels to safely divert river flow during dam and intake construction.

7. **Explain the specific consideration to be provided for the design, operation and management of Hill irrigation system. [10]**

Answer:

Steep slopes, rugged topography, unstable geology and scattered agricultural land makes the hill irrigation systems different from those in plains. Hence, specific considerations need to be taken for the design, operation and management of hill irrigation system. They are listed below:

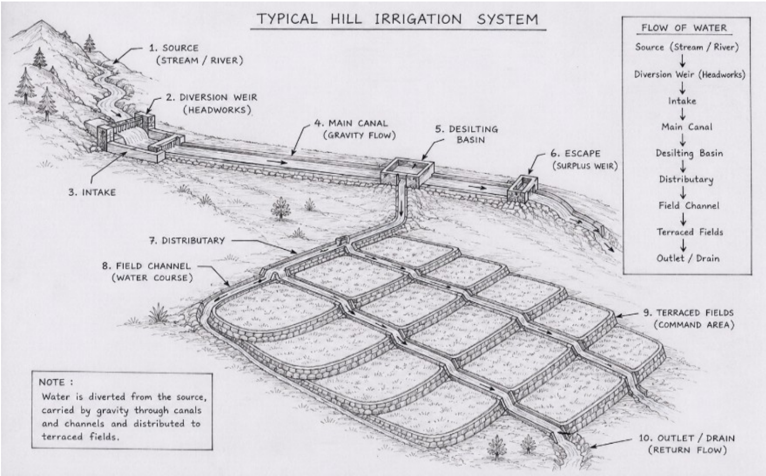
i. **Design considerations:**
   - Reliable source of water must be selected for the withdrawing adequate amount of discharge for irrigation systems.
   - Canals need to be aligned along the contour lines to minimize excavation and erosion.
   - Landslide prone areas, unstable slopes and excessive cutting of hillslopes must be avoided for construction of irrigation system.
   - Permissible flow velocity must be maintained in the irrigation systems to prevent erosion or sediment deposition.

ii. **Managerial and Institutional considerations:**
   - Regular inspections and maintenance work of canals, diversion structures, retaining walls etc. are required.
   - Regulating structures must be easy to use for the local farmers to operate and repair.

iii. **Agricultural considerations:**
   - The irrigation system design must be compatible with the soil type of the terrain and the crops that are grown in hilly terrain.
   - Water must be designed to be delivered in a way that are suitable to the local farming pattern.

iv. **Financial considerations:**
   - Design of the irrigation systems in the hill terrain must be economical and affordable.
   - Local labor and locally available materials should be used for construction, maintenance and repairs to improve long term sustainability and economical aspect.

v. **Social arrangements:**
   - Irrigation systems must be constructed by protecting the water rights, local customs and ethnic groups.
   - The construction of irrigation system should not instigate any conflict and make water distribution acceptable to users.



8. Write short notes on:

a. Farmer managed irrigation system [5]

b. Flood control, its necessary and mitigation measures [5]

Answer:

a.

- **Farmer Managed Irrigation Systems (FMIS)** are irrigation schemes that are developed, operated, and maintained entirely by the farming communities they serve. In the context of Nepal, these systems represent a significant part of the agricultural landscape and have a long historical tradition.
- Before 1922, irrigation in Nepal was primarily developed and managed through FMIS. It was only after this period that the government began making significant efforts to develop state-led irrigation infrastructure.
- FMIS are the backbone of Nepal's irrigated agriculture, covering approximately 70% of the country's total irrigated land.
- Some of the biggest FMIS maintained in Nepal are Rajapur Irrigation System, Rani Jamara Kulariya Irrigation System etc.
- Advantages of FMIS:
  i. Strong community participation and ownership.
  ii. Low operation and maintenance cost due to farmers' contributions.
  iii. Efficient and equitable water distribution based on local rules.
  iv. Quick repair and maintenance because users are directly involved.
  v. Uses local knowledge and indigenous technology.
  vi. Improves agricultural productivity and rural livelihoods.
  vii. Promotes social cooperation and conflict resolution among farmers.
  viii. Sustainable due to users' sense of responsibility.

b. Flood control refers to the structural and non-structural engineering measures designed to manage the flow of water, stabilize river channels, and prevent the inundation of agricultural land and human settlements. In the context of Nepal, which is the second richest country in water resources, flood control is particularly critical due to the steep slopes of the Himalayas and the dynamic nature of rivers in the Terai plains.

Flood control is essential in Nepal for several technical and environmental reasons:

- **Protection of Agricultural Land and Infrastructure:** Excess sediment originating from the Himalayas increases riverbed levels in the Terai, reducing channel capacity and significantly increasing the risk of flooding on agricultural land and nearby infrastructure.
- **Management of Dynamic River Morphology:** Rivers like the Koshi are highly prone to frequent flooding and lateral shifting. River training works are necessary to guide these waters within specific paths and protect settlements.
- **Climate Change and GLOFs:** Intensified and untimely rainfall due to climate change has increased the frequency of natural calamities. Furthermore, the melting of glaciers at increased rates creates the risk of Glacial Lake Outburst Floods (GLOFs), which can destroy downstream ecosystems and structures.
- **Infrastructure Integrity:** Proper flood estimation and management are required to design hydraulic structures such as dams, spillways, bridges, and culverts that can withstand peak discharge and scour.

Mitigation strategies are categorized into structural engineering works and non-structural management techniques.

- **Marginal Embankments (Levees):** Earthen embankments built parallel to riverbanks to confine floodwaters within a specific cross-section, particularly for meandering rivers.
- **Guide Banks:** Constructed at bridge sites to ensure the flow path does not change through the waterway, protecting the bridge foundation from shifting currents.
- **Flood Diversion Channels and Detention Basins:** These helps divert excess water away from high-risk areas or store it temporarily to reduce peak flow downstream.
- **Afforestation and Vegetation:** Planting trees and vegetative systems strengthens the soil mass, reduces the velocity of surface runoff, and facilitates infiltration and percolation into the groundwater table.
- **Managed Aquifer Recharge (MAR):** Directing stormwater and surface runoff into the ground recharges depleted aquifers while reducing surface flood volumes.

9. Note the type of foundation employed in a building constructed recently in your area. List and explain briefly obvious reasons why this type of foundation was selected. [10]

Answer:

The type of foundation selected depends upon soil type, amount of load to be subjected, type of building, topography, budget for construction and so on. The foundation employed in the building constructed recently in my area is generally shallow footing. Depending upon the type of building to be constructed, type of soil in the area and its bearing capacity, foundations can be selected. For example, if a commercial building is to constructed and the bearing capacity of the soil is relatively low, with soft soil, then mat footing is opted. For residential building subjected to lesser load, isolated footing can be provided.

In my area, foundation employed in a building constructed recently was isolated shallow footing. The footing is made of RCC and is provided with individual columns. This type of footing is suitable where the soil has adequate bearing capacity and the loads to be transferred to columns are moderate.

The reasons why this type of foundation was selected are listed below:

i. **Adequate Soil Bearing Capacity**:
Isolated footing is used when the construction area has firm soil capable of safely supporting the building load without excessive settlement.

ii. **Economical choice:**
This type of footing requires less excavation and reinforcement than raft or pile foundations, making it cost-effective.

iii. **Suitable for Low- to Medium-Rise Buildings**:
Isolated footing is appropriate for residential buildings of 2–5 storeys with moderate column loads.

iv. **Simple Construction**:
The construction of shallow foundation is easy to design and construct using locally available materials and labor.

v. **Low Groundwater Table**:
A shallow foundation can be constructed easily without extensive dewatering.

vi. **Uniform Soil Conditions**:
For the construction of shallow footing, the soil is relatively uniform, reducing the risk of differential settlement.

vii. **Adequate Spacing Between Columns**:
Isolated footing is preferred when the columns are suitably spaced so individual footings do not overlap.

viii. **Compliance with Building Codes**:
Properly designed isolated footings satisfy the requirements of the Nepal National Building Code (NBC) for such buildings.

10. What do you mean by participatory approach in planning implementation maintenance and operation of local infrastructures? Explain by giving examples. [10]

Answer:

A participatory approach is a process or the way in which local people, community organizations, local governments, and other stakeholders actively participate in the planning, design, implementation, operation, maintenance, monitoring, and decision-making of infrastructure projects. It ensures that the infrastructure meets the actual needs of the community ensuring sustainability and long-term utilization.

The different stages of participatory approach:

i. Planning:
Needs and priorities of the community is identified followed by the public meetings.
Project sites is selected on the basis of community consensus and mutual agreement.
The development plans in the participatory approach is made with public knowledge and resources.

ii. Implementation:
Participatory approach is characterized by the utilization of local raw materials, labors, cash etc.
Local users monitor the construction quality and mobilizes users' committee for monitoring.

iii. Operation:
User committee manages the day-day operation in participatory approach.
The rules and fees are established and collected by user's committee.

iv. Maintenance:
Routine maintenance is carried out by the users' groups.
Participatory approach ensures quick repairment of the damaged component through local participation.

Advantages of participatory approach in different stages of construction of local infrastructures are as follows:

- Creates a sense of ownership among users.
- Improves transparency and accountability.
- Reduces construction and maintenance costs.
- Increases project sustainability.
- Encourages participation of women and marginalized groups.

The examples of successful participatory approach in Nepal:

Community water supply scheme:

- Water Users and Sanitation Committees (WUSCs) plan, construct, operate, and maintain rural drinking water systems and also collect tariffs for maintenance and repairs.

Farmer managed irrigation system:

- FMIS are the backbone of Nepal's irrigated agriculture, covering approximately 70% of the country's total irrigated land.

Community forests:

- Forest User Groups protect and manage forests.
- Revenue from forest products is used for local infrastructure and development.

Construction projects through user committee:

- A **user committee** (often referred to as an **Upabhokta Samiti**) is a community-based body formed by local beneficiaries to actively participate in the planning, implementation, operation, and maintenance of projects.
- Active involvement of users themselves creates a strong sense of responsibility among users to protect and maintain the system.